

THE BINOMIAL EDGE JACOBIAN OF (K_3, K_3) LOSES RANK IN CHARACTERISTIC TWO

ABSTRACT. For graphs G on $[n]$ and H on $[m]$ let $J_{G,H} = (x_{i,k}x_{j,l} - x_{i,l}x_{j,k} \mid \{i, j\} \in E(G), \{k, l\} \in E(H))$ and let $\mathcal{H}(G, H)$ be the Jacobian of those generators. Landsittel and Nevo prove $\text{rk } \mathcal{H}(G, H) = \ell(J_{G,H})$ over a field of characteristic 0 or larger than 2^{nm} , and ask whether the characteristic hypothesis can be removed. It cannot. At $G = H = K_3$, where $\mathcal{H}$ is the Jacobian of the 3×3 adjugate, we exhibit the 9×9 matrix and evaluate it at $X = I$: over every field of characteristic 2, $\text{rk } \mathcal{H}(K_3, K_3) = 8$ while $\ell(J_{K_3, K_3}) = 9$; over every field of characteristic $\neq 2$ the two agree, with no bound on the characteristic at all. The verification is a 3×3 determinant. The spread half is already in the source: Landsittel and Nevo prove $\ell(J_{K_n, K_m}) = nm$ for $n, m \geq 3$ over an arbitrary field, and we reprove only the case used here, from adjugate identities. Together with the inequality $\text{rk } \mathcal{H} \leq \ell$ over an arbitrary field, which they quote from Beecken–Mittmann–Saxena, the rank computation is short.

1. THE QUESTION

Fix a field F , integers $n, m \geq 1$, the polynomial ring $S = F[x_{i,k} \mid i \in [n], k \in [m]]$ and its fraction field $K = F(x_{i,k})$. Write $X = (x_{i,k})$ for the generic $n \times m$ matrix. For graphs G on $[n]$ and H on $[m]$, the *binomial edge ideal of the pair* (G, H) is

$$J_{G,H} = (f_{e,e'} \mid e = \{i < j\} \in E(G), e' = \{k < l\} \in E(H)),$$

$$f_{e,e'} = x_{i,k}x_{j,l} - x_{i,l}x_{j,k},$$

and $\mathcal{H}(G, H) \in K^{e(G)e(H) \times nm}$ is the Jacobian $(\partial f_{e,e'} / \partial x_{i,k})$, its rows indexed by $E(G) \times E(H)$ and its columns by the variables. We write $\ell(I)$ for the analytic spread of I , the Krull dimension of its special fibre ring. Landsittel and Nevo [5] prove:

Proposition B. For any graphs G and H , the rank of the Jacobian $\mathcal{H}(G, H)$ of the minimal binomial generators of $J_{G,H}$ equals the analytic spread of $J_{G,H}$, over any field of characteristic zero or larger than $2^{|V(G)||V(H)|}$.

The threshold is inherited from [1, Theorem 8], which turns an algebraic matroid into a linear one in characteristic 0 or large enough. The authors ask whether it is an artefact:

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Question. Can the characteristic conditions in [their theorem on $J_G = J_{G, K_2}$] and Proposition B be removed? Is it true that for any field F , every algebraic matroid $F \subset F(g_1, \dots, g_N)$ defined by degree two forms g_i is representable?

Both statements are quoted from the e-print source of [5], in which the characteristic hypothesis is twice recorded as technical: the authors say that they have no counterexample in nonzero characteristic below 2^n and that the assumption is introduced in order to apply [1, Theorem 8], and later that possibly the restrictions can be removed. The bracketed phrase above replaces an internal cross-reference of that source. We quote rather than cite by number because the numbering of a published version was not available to us, so the statement being addressed here is fixed by the quotation and not by a number.

This note answers the Proposition B half of the Question, in both directions.

Theorem 1. *Let $G = H = K_3$, so that $n = m = 3$, X is the generic 3×3 matrix and $\mathcal{H} = \mathcal{H}(K_3, K_3)$ is the 9×9 matrix of Table 1. Then, over every field F ,*

$$\ell(J_{K_3, K_3}) = 9, \quad \text{rk}_K \mathcal{H} = \begin{cases} 9, & \text{char } F \neq 2, \\ 8, & \text{char } F = 2. \end{cases}$$

In particular the conclusion of Proposition B fails over every field of characteristic 2, so its characteristic hypothesis cannot be removed; and at this pair characteristic 2 is the only obstruction, the conclusion holding in every other characteristic with no bound whatsoever.

Nothing printed in [5] is contradicted. Proposition B as stated assumes $\text{char } F = 0$ or $\text{char } F > 2^9 = 512$, and $\mathbb{F}_2$ lies outside that hypothesis; what is refuted is the affirmative answer to the Question for Proposition B, which the authors pose as open. The degree-two representability half of the Question is untouched: at (K_3, K_3) over $\mathbb{F}_2$ the nine generators are algebraically independent (Lemma 3 below), so their algebraic matroid is the free matroid $U_{9,9}$, representable over every field. What fails is not representability of the matroid but the Jacobian's ability to compute it. Only the single pair (K_3, K_3) is treated below: nothing is claimed at any other pair, in either direction, and no failure of Proposition B's conclusion in odd characteristic is exhibited anywhere.

2. THE WITNESS

Order $E(K_3) = \{12, 13, 23\}$ and index the rows of $\mathcal{H}$ by (e, e') lexicographically with e' prioritised, following the notation of [5]; index the columns by $x_{1,1}, x_{1,2}, x_{1,3}, x_{2,1}, \dots, x_{3,3}$. Since $f_{e,e'} = x_{i,k}x_{j,l} - x_{i,l}x_{j,k}$ has exactly four terms of degree 1 in each variable it involves, each row of $\mathcal{H}$ has exactly four nonzero entries, each a signed variable:

Table 1. $\mathcal{H}(K_3, K_3)$.

(e, e')	x_{11}	x_{12}	x_{13}	x_{21}	x_{22}	x_{23}	x_{31}	x_{32}	x_{33}
(12, 12)	x_{22}	$-x_{21}$	0	$-x_{12}$	x_{11}	0	0	0	0
(13, 12)	x_{32}	$-x_{31}$	0	0	0	0	$-x_{12}$	x_{11}	0
(23, 12)	0	0	0	x_{32}	$-x_{31}$	0	$-x_{22}$	x_{21}	0
(12, 13)	x_{23}	0	$-x_{21}$	$-x_{13}$	0	x_{11}	0	0	0
(13, 13)	x_{33}	0	$-x_{31}$	0	0	0	$-x_{13}$	0	x_{11}
(23, 13)	0	0	0	x_{33}	0	$-x_{31}$	$-x_{23}$	0	x_{21}
(12, 23)	0	x_{23}	$-x_{22}$	0	$-x_{13}$	x_{12}	0	0	0
(13, 23)	0	x_{33}	$-x_{32}$	0	0	0	0	$-x_{13}$	x_{12}
(23, 23)	0	0	0	0	x_{33}	$-x_{32}$	0	$-x_{23}$	x_{22}

The nine generators are exactly the nine 2×2 minors of X , that is, up to sign the nine entries of $\text{adj } X$; so $\mathcal{H}$ is the Jacobian of the adjugate map, equivalently the Hessian of the cubic form $\det_3$.

Specialising Table 1 at $X = I$ gives an integer matrix that a reader can put in echelon form by eye.

Table 2. $\mathcal{H}(K_3, K_3)$ at $X = I$.

(e, e')	x_{11}	x_{12}	x_{13}	x_{21}	x_{22}	x_{23}	x_{31}	x_{32}	x_{33}
(12, 12)	1	0	0	0	1	0	0	0	0
(13, 12)	0	0	0	0	0	0	0	1	0
(23, 12)	0	0	0	0	0	0	-1	0	0
(12, 13)	0	0	0	0	0	1	0	0	0
(13, 13)	1	0	0	0	0	0	0	0	1
(23, 13)	0	0	0	1	0	0	0	0	0
(12, 23)	0	0	-1	0	0	0	0	0	0
(13, 23)	0	1	0	0	0	0	0	0	0
(23, 23)	0	0	0	0	1	0	0	0	1

Six of the nine rows of Table 2 — all but (12, 12), (13, 13), (23, 23) — carry a single ± 1 , and those six entries sit in six *distinct* columns, namely $x_{32}, x_{31}, x_{23}, x_{21}, x_{13}, x_{12}$. Deleting those six rows and columns leaves the three surviving rows supported on x_{11}, x_{22}, x_{33} , where they read

$$(1) \quad A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \quad \det A = -2.$$

So $\text{rk } \mathcal{H}(I) = 6 + \text{rk } A$ over any field, and $\det \mathcal{H}(I) = \pm 2$.

Lemma 2. *Let G, H be any graphs and let $v = (x_{i,k}) \in K^{nm}$ be the Euler vector. Then $\mathcal{H}(G, H)v = 2f$, where f is the column of generators. Consequently, if $\text{char } F = 2$ then v is a nonzero kernel vector and $\text{rk}_K \mathcal{H}(G, H) \leq nm - 1$.*

Proof. Every $f_{e,e'}$ is homogeneous of degree 2, so Euler's identity gives $\sum_{i,k} x_{i,k} \partial f_{e,e'} / \partial x_{i,k} = 2f_{e,e'}$, which is the (e, e') entry of $\mathcal{H}v$. If $2 = 0$ in F then $\mathcal{H}v = 0$, and $v \neq 0$. $\square$

Lemma 3. *Over every field F , $\ell(J_{K_3, K_3}) = 9$; equivalently the nine 2×2 minors of the generic 3×3 matrix are algebraically independent over F .*

Proof. Let $g_1, \dots, g_9$ be the generators and $L = F(g_1, \dots, g_9) \subset K$. They are forms of the common degree 2, so the special fibre ring of J_{K_3, K_3} is $F[g_1, \dots, g_9]$ and $\ell(J_{K_3, K_3}) = \text{trdeg}_F L$ by the equigenerated-ideal lemma of [5], which carries no hypothesis on F and is taken there from [3, Proposition 4.8]. The entries of $\text{adj } X$ are, up to sign, the g_i , so they lie in L ; and the classical identities over $\mathbb{Z}$

$$\text{adj}(\text{adj } X) = \det(X) X, \quad \det(\text{adj } X) = \det(X)^2$$

give, for each of the nine entries,

$$x_{i,k}^2 = \frac{(\text{adj}(\text{adj } X)_{i,k})^2}{\det(\text{adj } X)} \in L.$$

Hence $K^2 \subset L \subset K$, so K/L is algebraic and $\text{trdeg}_F L = \text{trdeg}_F K = 9$. $\square$

Proof of Theorem 1. Lemma 3 gives $\ell(J_{K_3, K_3}) = 9$ over every field. For the rank, work with Table 2 and (1).

If $\text{char } F \neq 2$ then $\det A = -2 \neq 0$ in F , so $\text{rk } \mathcal{H}(I) = 6 + 3 = 9$ by the paragraph after Table 2; a specialisation cannot raise rank, so $9 = \text{rk } \mathcal{H}(I) \leq \text{rk}_K \mathcal{H} \leq 9$, the last bound being the number of columns. Hence $\text{rk}_K \mathcal{H} = 9 = \ell(J_{K_3, K_3})$: the conclusion of Proposition B holds at (K_3, K_3) over every field of characteristic $\neq 2$, with no threshold.

If $\text{char } F = 2$ then the three rows of A sum to zero, and any two of them are independent, so $\text{rk } A = 2$ and $\text{rk } \mathcal{H}(I) = 6 + 2 = 8$; thus $\text{rk}_K \mathcal{H} \geq 8$. Lemma 2 gives $\text{rk}_K \mathcal{H} \leq nm - 1 = 8$. Hence $\text{rk}_K \mathcal{H} = 8 < 9 = \ell(J_{K_3, K_3})$. $\square$

Remark. The kernel of $\mathcal{H}(I)$ over a field of characteristic 2 is exactly the line spanned by $(1, 0, 0, 0, 1, 0, 0, 0, 1)$, which is the Euler vector $v = (x_{i,k})$ evaluated at $X = I$: the two upper bounds of the proof are the same vector seen twice, and this is why the deficiency is exactly 1 and not more.

Remark. The determinant of Table 1 is $\det \mathcal{H}(K_3, K_3) = 2 \det(X)^3$, an identity in $\mathbb{Z}[x_{1,1}, \dots, x_{3,3}]$ with 54 monomials on each side. (The *sign* depends on the chosen orders of the rows and of the columns and is therefore not an invariant; the integer content 2 is, and it is all that is used.) Nothing about this object is new: it is the case $n = 3$ of the classical evaluation of the Hessian of $\det_n$ as $(-1)^{n^2-1}(n-1) \det(X)^{n(n-2)}$, whose integer factor at $n = 3$ is $n - 1 = 2$.

3. PRIOR ART AND ATTRIBUTION

The ingredients are other people's. The inequality $\text{rk}_K \mathcal{H}(G, H) \leq \ell(J_{G,H})$ over an arbitrary field is [1, Lemma 9], quoted as such in [5]. The phenomenon that an algebraic matroid over a field of positive characteristic need not be representable is Ingleton's [4, p. 166, Example 15], also cited by [5]. The

determinant identity in the last remark of Section 2 is the classical Hessian of $\det_3$.

The spread half of Theorem 1 is in the cited source. Landsittel and Nevo prove $\ell(J_{K_n, K_m}) = nm$ for $n, m \geq 3$ and remark explicitly that the proof needs no assumption on the field and makes no reference to representability; at $n = m = 3$ that statement is Lemma 3. Their proof runs through [2, Theorem 1], which we did not check for field generality, and that is the only reason Lemma 3 is proved here from the adjugate identities; the value is theirs. Granted that value and [1, Lemma 9], what remains of the refutation is the Euler vector of Lemma 2 together with the evaluation in Table 2.

A search for the rank statement itself returned nothing containing it, and we make no priority claim on that basis. The nearest items found do not settle it: [5] record it as open twice; [1] bounds $\text{rk } \mathcal{H}$ by ℓ but evaluates neither at any pair; [2] inverts the multiplicative compound operator and treats neither the Jacobian rank nor characteristic 2; and Ingleton's example produces a non-representable algebraic matroid, whereas at (K_3, K_3) over $\mathbb{F}_2$ the matroid is free and what fails is the Jacobian.

4. VERIFICATION

Everything decisive is in Tables 1 and 2, (1) and Lemmas 2–3: a referee who wants to run nothing needs one 3×3 determinant, $\det A = -2$, plus the observation that the rows of A sum to zero modulo 2.

The accompanying program `verify.py` (Python 3.9+, standard library only, no external data file) re-derives from the printed edge lists, rather than reading, the quantities used above: it rebuilds the nine generators and checks they are the nine 2×2 minors and the signed entries of $\text{adj } X$; reproduces Table 1 entry by entry against the symbolic Jacobian; expands $\det \mathcal{H} = 2 \det(X)^3$ exactly over $\mathbb{Z}$ and reports its content 2 and its vanishing modulo 2; verifies $\mathcal{H}v = 2f$ and the adjugate identities of Lemma 3 as polynomial identities over $\mathbb{Z}$; and recomputes the ranks at $X = I$ over $\mathbb{Q}$, over $\mathbb{F}_2$ (with the kernel) and over $\mathbb{F}_p$ for $p = 3, 5, 7, 11, 13, 509$. All arithmetic is exact: integers, `fractions.Fraction`, and a $GF(2^{16})$ whose modulus is proved irreducible inside the program; every characteristic-2 rank it reports is an evaluation, hence a lower bound, and is asserted only once it meets a proved upper bound. The recorded run `verify.output.txt` reports 75 checks, all passing.

That run also carries checks for statements this note does not make: a family of pairs whose vertices lie in chained triangles, the pair $(2K_3, 2K_3)$, and the (K_n, K_2) side of the source's other theorem. Nothing above rests on them. One of those lines is labelled as saying that characteristic 2 degenerates only at $n = 3$, which is false as stated, since 2 divides $n - 1$ for every odd n ; no claim in this note depends on it.

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